COVER LETTER

Date: 28/04/2026

To,

The Editor,

Knowledge and Information Systems

Subject: Submission of Manuscript Titled **"Reliability-Aware Biomedical Named Entity Recognition via Dual-Path Extraction and Confidence Calibration"**

Dear Editor,

I am submitting our manuscript, "Reliability-Aware Biomedical Named Entity Recognition via Dual-Path Extraction and Confidence Calibration," for consideration in your journal.

Context and Importance

Biomedical named entity recognition is essential for extracting structured knowledge from biomedical text, yet existing models often lack reliable confidence estimation, limiting their use in safety-critical applications. To address this, we propose a reliability-aware framework that integrates dual-path extraction using both large language models and domain-specific models, combined with candidate fusion, reflective refinement, and calibrated confidence estimation. Our approach enables selective prediction by balancing precision and coverage, and experiments on BC5CDR and NCBI Disease datasets demonstrate improved reliability and clearer distinction between trustworthy and uncertain predictions.

Appropriateness for the Journal

The manuscript aligns well with the scope of *Knowledge and Information Systems*, as it addresses key challenges in building intelligent systems that integrate multiple knowledge sources, reasoning mechanisms, and reliability-aware decision processes. The proposed framework contributes to the broader goal of developing trustworthy and interpretable information extraction systems, particularly in domains where reliability is as critical as accuracy. The combination of multi-agent reasoning, dual-path modeling, and calibrated confidence estimation reflects the journal's focus on advanced knowledge-driven systems and their practical applications.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have reviewed and approved the manuscript and agree with its submission to your journal.

Thank you for your time and consideration.

Sincerely,

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